PNAS RESEARCH ARTICLE | CLASSIFICATION

Finite-time Lyapunov exponents in conservative systems: the importance of rare compression events

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This manuscript was compiled on October 7, 2026

We demonstrate that, in conservative dynamical systems, rare compression events dictate the shape of the distribution of finite-time Lyapunov exponents. In particular, we show that its low-stretch tail is exponential, a result supported by four examples: stagnation points, sine flows, homogeneous isotropic turbulence and an experimental chaotic flow in a rotating cylinder. This finding has important consequences for scalar mixing in hydrodynamic flows. Weakly stretched fluid parcels aligned with the compression direction have long-lived concentration levels c that dominate the scalar concentration distributions at late time and build up a persistent c^{-3} tail, in all space dimensions.

Lyapunov exponents | Turbulence | Mixing

Significance

Numerous phenomena in the tangible world are not governed by a mean but rather by significant deviations from it. For instance, odors are perceived through yet unmixed parcels of carrying fluid remained intact far from their source; delayed spray evaporation facilitates long-distance disease transmission from human exhalations, among others. Because of a geometric constraint due to incompressibility, the tail of the low Lyapunov exponents distribution (which quantify mixing rates) are exponentially distributed, implying that large, practically unaltered concentration levels from the source are algebraically distributed in stirred mixtures, in all dimensions of space. In chaotic and turbulent flows, these rare events have a persistent consequence on the mixture composition. Exceptions are, in fact, the rule.

Lyapunov exponents characterize the degree of chaoticity in dynamical systems; they quantify the divergence rate of initially nearby trajectories (1). In a dynamical system $\dot{\boldsymbol{X}} = \boldsymbol{u}(\boldsymbol{X})$, two trajectories $\boldsymbol{X}(t)$ and $\boldsymbol{X}'(t)$ initially separated by $\boldsymbol{\ell}_0 = \boldsymbol{X}_0 - \boldsymbol{X}'_0$ are separated at a later time t by a distance $\ell = \sqrt{{}^T\boldsymbol{\ell}_0 {}^T Y Y \boldsymbol{\ell}_0}$ with Y the Jacobian matrix of the flow $\boldsymbol{u}$ satisfying $\dot{Y} = (\nabla \boldsymbol{u})Y$ and $Y(t=0) = I$ is the identity matrix. A *finite-time* Lyapunov exponent is then defined from the elongation $\rho = \ell/\ell_0$ as

$$\lambda = \frac{1}{t} \ln \rho. \qquad [1]$$

At a given instant of time, ρ and λ depend on the initial position $\boldsymbol{X}_0$ of the trajectory but also on the initial separation $\boldsymbol{\ell}_0$. They are hence distributed around a mean value.

Let us first consider initial separation $\boldsymbol{\ell}_0$ aligned with the stretching and compression directions on $\boldsymbol{u}$, i.e. the eigenvectors $\boldsymbol{e}_1, \boldsymbol{e}_2, \cdots$ of the matrix ${}^T Y Y$. By definition, the squared elongations $\rho_1^2 \geq \rho_2^2 \geq \cdots$ are the eigenvalues of ${}^T Y Y$ and $\lambda_1 \geq \lambda_2 \geq \cdots$ are the associated finite-time Lyapunov exponents. For $\lambda_i < 0$, the direction $\boldsymbol{e}_i$ is compressing and for $\lambda_i > 0$, it is stretching; for $\lambda_i = 0$, the direction $\boldsymbol{e}_i$ is neutral. The exponents λ_i do not depend on the initial orientation of $\boldsymbol{\ell}_0$ anymore, but only on the initial position $\boldsymbol{X}_0$. Their distribution converges for asymptotically long times towards Dirac distributions (2), and the rate of convergence has been studied in (3), for instance. Here, we focus on finite-time effects.

An orientation of $\boldsymbol{\ell}_0$ involving an amount of compression (i.e. a direction $\boldsymbol{e}_i$ with $\lambda_i < 0$) leads to Lyapunov exponents much smaller than the average one. When applied to a passive scalar in a flow, this effect will delay stretching-enhanced diffusion (4), with consequences on the mixture composition. It is hence crucial to consider the full distribution of λ, since it takes into account all possible orientation of the initial separation $\boldsymbol{\ell}_0$ with respect to compression and stretching directions. Here, we quantify the importance of rare (in the sense that they correspond to a particular initial orientation) compression events in the probability density function (p.d.f.) of the Lyapunov exponent λ, and their impact on the scalar statistics.

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The theory has been developed by NP, PM and EV, the numerical simulations were performed by NP, SMBK and PM, the 3D experiment was done by PM. The paper was written by NP, PM and EV.

The authors report no conflict of interest.

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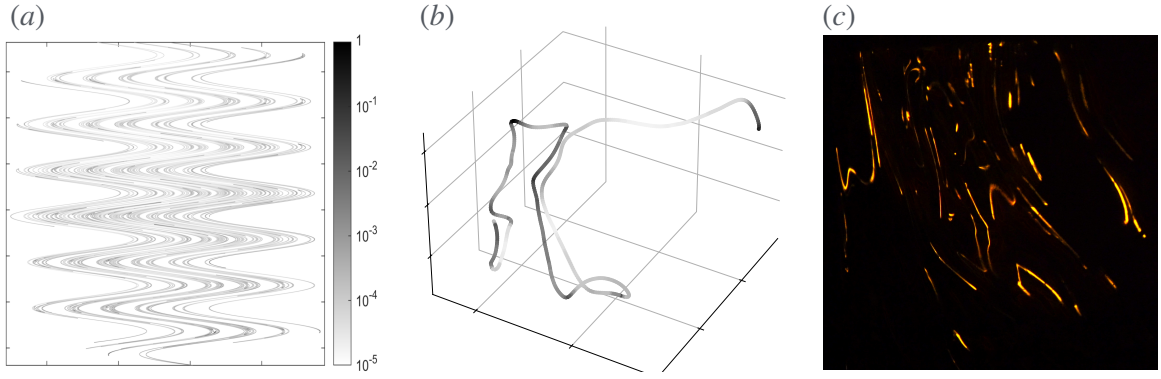


Fig. 1. Illustration of the concentration fields built from scalar carrying lamellae stretched by the different stirring protocols studied here: A 2D Sine flow (a), a 3D turbulent flow (b), and an experimental 3D chaotic flow (c). Note the omnipresence of tilted, reorienting lamellae, with concentration much larger than the mean.

In chaotic systems, it is often assumed that the central limit theorem holds and that Lyapunov exponents should exhibit Gaussian statistics (5), with mean and variance related to the dimensionality of space (6, 7). This has been proved analytically for random flows delta-correlated in time (7) and measured in direct numerical simulations of homogeneous isotropic turbulence (8), although in that case the Gaussian p.d.f. were shown to be slightly skewed (see also (9)).

For *dissipative* systems with $\nabla \cdot \boldsymbol{u} \neq 0$, non-Gaussian deviations have been observed. The logistic map displays Lyapunov exponents with (stretched) exponential tails (2, 10). Similar results have been predicted for large-dimensional linear stochastic systems (11).

In *conservative* systems, such as incompressible fluid flows and turbulence, the constrain $\nabla \cdot \boldsymbol{u} = 0$ imposes $\rho_1\rho_2 \cdots = 1$ or, equivalently, $\lambda_1 + \lambda_2 + \cdots = 0$. In other words, stretching in a direction is balanced by compression in another. Emphasis have been put on the analysis of the largest finite-time Lyapunov exponent λ_1. Particular examples (such as the Hénon-Heiles model) have been studied (12, 13), reporting strong non-Gaussianity of the distribution of λ_1. Non-turbulent conservative chaotic flows have also been found to exhibit non-Gaussian (almost exponential) high-stretch (large λ_1) tails (14). Three-dimensional turbulence has also been analysed; in a seminal paper, Johnson and Meneveau (15) studied the structure of the distribution of both λ_1, λ_2 and λ_3 using large deviation theory.

Here, we focus on the distribution of λ, taking into account all the possible orientations of the initial separation $\boldsymbol{\ell}_0$ with respect to both the stretching and compression directions. Especially, we focus on its on the low-stretch (small λ) tails because it controls the large concentration levels in mixing problems (mixing times scale as λ^{-1} and elements stretched at large λ have a vanishingly small concentration (16)). Do Lyapunov exponents follow the central limit theorem, or do rare—but inevitable, due to incompressibility—compression events pollute the statistics in a significant manner? We show that the latter claim is generically true, and derive its consequences for scalar mixing in a broad range of flows.

Distribution of Lyapunov exponents

We characterize departures from Gaussian statistics in systems with increasing complexity, including chaotic, and turbulent flows, in both two and three dimensions (Fig. 1). We first set the scene with a simple but instructive example.

Two-dimensional stagnation points. Up to translation and rotation, two-dimensional incompressible flows can *locally* be approximated by stagnation points (Fig. 2) of the form $\boldsymbol{u} = {}^T[-\gamma x, \gamma y]$, with a compression direction aligned with the x-axis and a stretching direction aligned with the y-axis and $\lambda_1 = \gamma$, $\lambda_2 = -\gamma$. Considering an initial separation $\boldsymbol{\ell}_0 = {}^T[\ell_0 \cos\theta, \ell_0 \sin\theta]$, we have

$$\lambda = \frac{1}{2t} \ln\left(\mathrm{e}^{-2\gamma t}\cos^2\theta + \mathrm{e}^{2\gamma t}\sin^2\theta\right) \underset{\gamma t \gg 1,\ \theta \neq 0}{\sim} \gamma + \frac{\ln(\sin\theta)}{t}. \qquad [2]$$

At late times, all Lyapunov exponents converge toward γ, corresponding to the stretching direction $\theta = \pi/2$ (unless the initial separation $\boldsymbol{\ell}_0$ is aligned with the compression direction $\theta = 0$; in that case, $\lambda = \lambda_2$ at all times). At finite times, some directions close to $\theta = 0$ are initially compressed, with smaller Lyapunov exponents. Sampling over a uniform distribution $h(\theta) = 2/\pi$ of initial orientations $\theta \in [0, \pi/2]$, the conservation of probability $q(\lambda)\mathrm{d}\lambda = h(\theta)\mathrm{d}\theta$ gives

$$q(\lambda)\left|\frac{\cos\theta}{t\sin\theta}\right| = \frac{2}{\pi}. \qquad [3]$$

The distribution of Lyapunov exponent $q(\lambda)$ is found by writing $\sin\theta$ as a function of λ from Eq. (2). Close to the compression direction in $\theta = 0$ where $\cos\theta \simeq 1$,

$$q(\lambda) \simeq \frac{2t}{\pi}\mathrm{e}^{(\lambda-\gamma)t}. \qquad [4]$$

At a stagnation point, the low-stretch tail (for $\lambda < \gamma$) of the distribution of Lyapunov exponents is *exponential*. Exact formulae for $q(\lambda)$ (given in the Exact formulae section) are plotted as colored lines in Fig. 2, exhibiting the exponential tails bounded by divergences at $\pm\gamma$. The corresponding uniform elongations distribution $Q(\rho) \sim \mathrm{e}^{-\gamma t}$ complies to the incompressibility constraint.

Extension to three dimensions. The above argument can be extended for a three-dimensional stagnation point $\boldsymbol{u} = {}^T[-\gamma x, 0, \gamma z]$. The stretching direction is now aligned with the z-axis, while the y-direction is neutral (Fig. 3),

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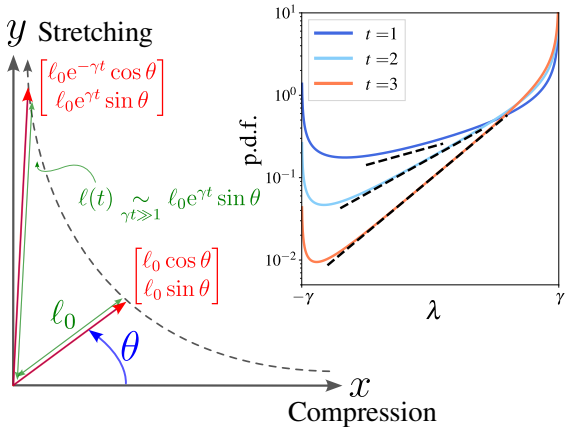


Fig. 2. Stretching in a two-dimensional stagnation point. The dashed line corresponds to a streamline associated with the velocity field $\boldsymbol{u} = {}^T[-\gamma x, \gamma y]$. The probability distribution of Lyapunov exponents is presented in the inset for three dimensionless times t, for $\gamma = 1$ (solid coloured lines). The dashed lines correspond to the approximation $2te^{\lambda t-\gamma t}/\pi$ in Eq. (4).

and $\lambda_1 = \gamma$, $\lambda_2 = 0$, and $\lambda_3 = -\gamma$. In that case, we parameterize the initial condition using the spherical angles θ and ϕ as $\boldsymbol{\ell}_0 = {}^T[\ell_0 \sin\theta\cos\phi, \ell_0 \sin\theta\sin\phi, \ell_0\cos\theta]$. In that case,

$$\lambda = \frac{1}{2t}\ln\left(\sin^2(\theta)\cos^2(\phi)e^{-2\gamma t} + \cos^2(\theta)e^{2\gamma t}\right). \quad [5]$$

At large times $\gamma t \gg 1$, we have

$$\lambda \simeq \gamma + \frac{\ln(\cos\theta)}{t} \xrightarrow[\gamma t\gg 1]{} \gamma, \quad [6]$$

unless $\theta = \pi/2$, i.e. the initial condition is in the neutral-compression plane; then, the λ converges towards the neutral Lyapunov exponents $\lambda_2 = 0$ if $\phi \neq 0$ or the compression one $\lambda_3 = -\gamma$ if $\phi = 0$. Since at late times λ is independent of ϕ, the probability $q(\lambda)\mathrm{d}\lambda$ equals the probability $h(\theta)\mathrm{d}\theta$ of sampling an initial orientation with colatitude within $[\theta, \theta + \mathrm{d}\theta]$

$$q(\lambda)\mathrm{d}\lambda = h(\theta)\mathrm{d}\theta = \frac{\int_{\phi=0}^{\pi/2}\sin\theta\mathrm{d}\phi\mathrm{d}\theta}{\pi/2} = \sin\theta\mathrm{d}\theta, \quad [7]$$

so that, after differentiating Eq. (6),

$$q(\lambda) \simeq t\cos\theta \simeq te^{(\lambda-\gamma)t}. \quad [8]$$

Exact formulae (derived in the Exact formulae section) show that the approximation above is valid for λ in between $\lambda_1 = \gamma$ and $\lambda_2 = 0$, corresponding to the Lyapunov exponents for initial conditions $\boldsymbol{\ell}_0$ aligned with the stretching and neutral direction, respectively.

Deformation of surface elements are analyzed along the same procedure by considering $\dot{\boldsymbol{\ell}} = -\left({}^T\nabla\boldsymbol{u}\right)\boldsymbol{\ell}$ rather than $\dot{\boldsymbol{\ell}} = (\nabla\boldsymbol{u})\boldsymbol{\ell}$ (17). In that case, $\boldsymbol{\ell}$ is a vector normal to surface elements with norm equal to their area (Fig. 3(*b*)). Up to finding $\boldsymbol{v}$ such that $\nabla\boldsymbol{v} = -{}^T\nabla\boldsymbol{u}$, the line and surface stretching problems are equivalent. In the stagnation point considered here, this comes down to considering $\boldsymbol{v} = {}^T[\gamma x, 0, -\gamma z] = -\boldsymbol{u}$. The p.d.f.s are thus equal.

The above predictions, derived in idealized flows, are now compared with numerical results in more elaborate situations, namely a chaotic two-dimensional flow (Fig. 1(a)), three-dimensional turbulence (Fig. 1(b)), and an experiment involving a three-dimensional chaotic flow (Fig. 1(c)). Throughout the rest of this work, the distribution of λ are computed based on the distribution of λ_1, λ_2, and λ_3 using the diffuselet method developed in (17).

Chaotic flows. We first consider a sine flow (in two dimensions). It is defined at the n-th interval of time, as

$$\begin{aligned}\boldsymbol{u} &= U\begin{bmatrix}0\\ \sin(2\pi x + \chi_1(n))\end{bmatrix}, \text{ if } t\in[n, n+1/2],\\ \boldsymbol{u} &= U\begin{bmatrix}\sin(2\pi y + \chi_2(n))\\ 0\end{bmatrix}, \text{ if } t\in[n+1/2, n+1].\end{aligned} \quad [9]$$

Here $U = 3$ is the amplitude of the flow and χ_1 and χ_2 are random phases. This flow and its stretching statistics have been extensively studied in (17). In the previous cases, the Lyapunov exponent in the stretching direction was fixed at a value $\lambda_1 = \gamma$. This maximal Lyapunov exponent now depends on the initial position $\boldsymbol{X}_0$ and is thus distributed around a mean value. These distributions of maximal Lyapunov exponents $\lambda_1 > 0$ (i.e. for the stretching direction) are plotted as triangles in Fig. 4(*a*). They look like Gaussians getting narrower in time, in agreement with the central limit theorem. However, for each initial position $\boldsymbol{X}_0$, the p.d.f. $q(\lambda)$ contains additional information from the variation of the initial orientation of $\boldsymbol{\ell}_0$ with respect to the stretching and compression directions, that, according to the above, translates into exponential tails. Fig. 4(*a*), obtained by sampling uniformly over initial orientations, shows that $\lambda^{-1}\ln q$ has slope equal to t (solid black lines); the low-stretched exponential tails described before in an idealized flow are present in a heterogeneous chaotic flow. This exponential tail converges toward the final Dirac delta like $1/t$, which is consistent with the rate of convergence of the so-called 'instability exponents' in (3).

Homogeneous isotropic turbulence. Similar results are observed in homogeneous isotropic turbulence for both lines and surfaces (Figs 4(*b*) and 4(*c*)). Here, we used simulation R1 from (19), a forced direct numerical simulation of homogeneous isotropic turbulence with Taylor Reynolds number $Re_\lambda \simeq 102$. The elongation statistics $Q(\rho)$ are determined using formulae (2.46-2.48) of (17). They require computing the eigenvalues $\rho_1^2 \geq \rho_2^2 \geq \rho_3^2$ of ${}^TYY$ along multiple trajectories $\dot{X} = \boldsymbol{u}(X)$. Here we considered 10000 trajectories with initial positions uniformly sampled in the simulation box and initial orientations also sampled uniformly. To ensure the incompressibility condition $\rho_1\rho_2\rho_3 = 1$ at all times, the eigenvalues are computed using the method described in (3) [see also Materials and Methods section]. The distribution of Lyapunov exponents follows from the change of variable $\rho = e^{\lambda t}$.

Consequences for mixing

A proxy of $q(\lambda)$ in a flow carrying a passive scalar is the distribution $p(c)$ of the concentration c transported by the stretched elements. Because a mixture can be viewed as an ensemble of lamellae whose scalar content evolves under the action of substrate deformation and molecular diffusivity (16), the above results have indeed important

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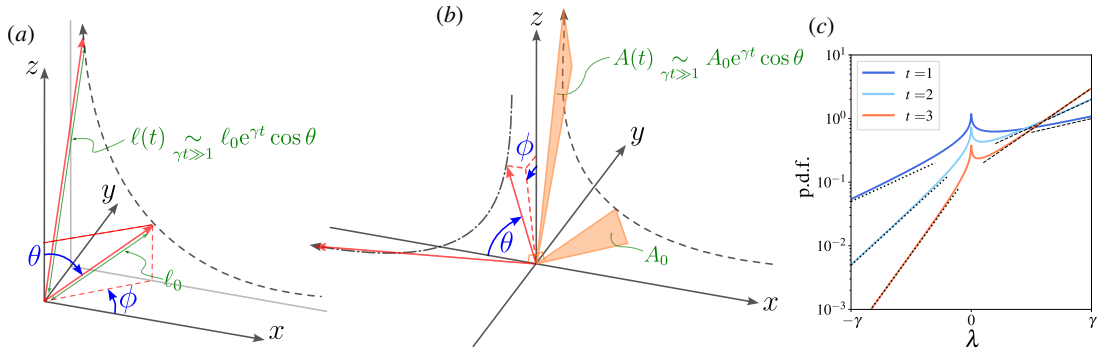


Fig. 3. Line (*a*) and surface (*b*) stretching in a three-dimensional stagnation point (the black dashed line corresponds to a streamline of $\boldsymbol{u}$). The cases of surfaces and lines are equivalent if one considers $\boldsymbol{v} = -\boldsymbol{u}$, or equivalently, swap the stretching and compression directions (the black dash-dotted line corresponds to a streamline of $\boldsymbol{v}$). In that case, the (red) vector $\boldsymbol{\ell}$ is normal to the stretching surface with norm given by its area (17). (*c*) Distribution of Lyapunov exponents in the stagnation point for three dimensionless times t, with $\gamma = 1$ (coloured lines). The dashed and dotted lines correspond to the asymptotes $q(\lambda) \simeq te^{\lambda t - \gamma t}$ and $q(\lambda) \simeq te^{2\lambda t - \gamma t}$ at $\lambda \simeq \gamma$ and $\lambda \simeq -\gamma$, respectively (see Exact formulae section).

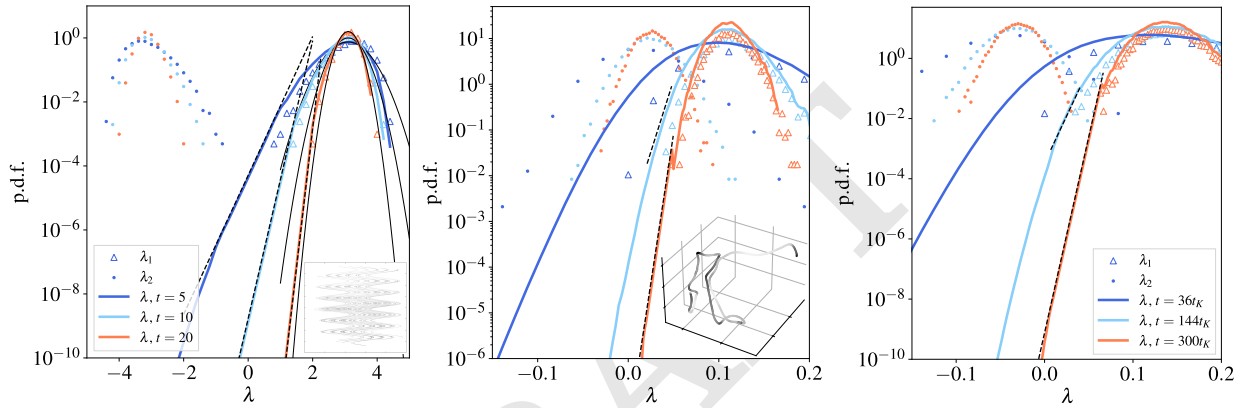


Fig. 4. (*a*) Distribution of Lyapunov exponents in the sine flow in Eq. (9) [adapted from (17)]. Thick colored lines correspond to the distribution of finite-time Lyapunov exponents $q(\lambda)$, at different times; these lines are not fits to the data and are computed using formula (2.46) of (17). Solid black lines correspond to Gaussian fits to these distributions. Triangles and dots represent the distribution of the Lyapunov exponents $\lambda_1 > 0$ and λ_2 for initial condition $\boldsymbol{\ell}_0$ aligned with the stretching and neutral direction, respectively. Note that while λ_1 is always non-negative, λ can be both positive and negative since it takes into account all possible orientations of the initial separation $\boldsymbol{\ell}_0$ with respect to stretching and compressing directions. (*b*) Distribution of Lyapunov exponents in homogeneous isotropic turbulence for lines, scaled by the inverse of the Kolmogorov time-scale t_K. Note that the most probable values of λ_1, λ_2 and λ_3 (scaled by t_K^{-1}) are 0.113, 0.026 and -0.140, respectively, consistent with the values of 0.114, 0.029 and -0.143 of Johnson and Meneveau (15). (*c*) Same for surfaces. In the three panels, the black dashed lines have slope t. Deforming lines in the insets in panel (*a*) and (*b*) are coloured based on the local value of elongation, with darker regions corresponding to less elongated ones.

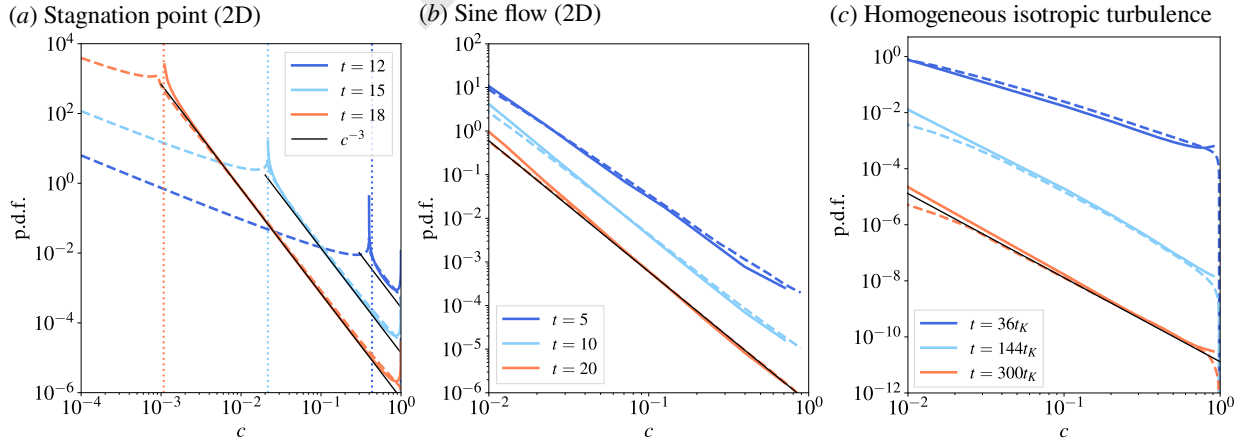


Fig. 5. Concentration distribution for (*a*) the stagnation point with $Pe = 10^{10}$, (*b*) the sine flow with $Pe = 3 \times 10^6$ and (*c*) homogeneous isotropic turbulence with $Pe = 10^3$. Solid colored lines correspond to the p.d.f of the maximal concentration along lamellae $p(c)$. Dashed ones correspond to the total concentration p.d.f. $P(C)$ for Gaussian concentration profiles given in Eq. (25). In (*a*), vertical dotted lines correspond to the density level $c_{\text{min}} = e^{-\gamma t}\sqrt{Pe/2}$. At early times in the sine flow and turbulent cases, the low-stretch exponential tail of $q(\lambda)$ has not yet formed (Fig. 4) and hence the c^{-3} scaling law is not yet observed. The Péclet number $Pe = \gamma s_0^2/D$ is defined using the shear: in (*b*) $\gamma = U = 3$ since the characteristic length scale is equal to 1, $s_0 = 10^{-4}$ and $D = 10^{-14}$; in (*c*) $\gamma \simeq 0.2/t_K$ is the mean Lyapunov exponent (with 0.2 the approximate average of the distribution of λ presented in Fig. 4(*c*)), $s_0 = 10^{-3}$ and $D = 10^{-7}$.

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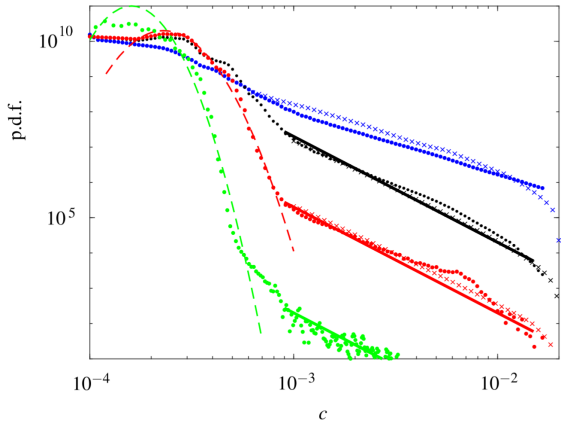


Fig. 6. Concentration distribution of a fluorescent dye (D =) in a soft-mixer (18) rotating at a speed of and title with an angle of, after 30 (blue), 70 (black), 110 (red) and 150 (green) rotation periods. The dots correspond to experimental measurements and the crosses to a numerical reconstruction of the concentration field using the diffuselet method (17). Dash lines correspond to Gaussian fits and thick coloured lines to the $p(c) \sim c^{-3}$ prediction.

consequences for mixing in incompressible flows. In a random mixture, the lamellae may not all have the same orientation with respect to the stretching direction of the substrate. We demonstrated that directions aligned along the compression axis have a significant impact on the deformation statistics. How is this imprint translating into the evolution of the scalar content in a mixture? We show next that it is directly responsible for the existence of long-lived, large concentration levels in mixtures.

On a still substrate, a lamella with a Gaussian initial concentration profile $C(x) = c_0 \exp(-x^2/s_0^2)$ diffuses such that its maximum concentration evolves as

$$c = \frac{c_0}{\sqrt{1 + \frac{4D}{s_0^2}t}}, \qquad [10]$$

with s_0 the initial lamella width, c_0 the initial maximal concentration and D the diffusivity. On a stirred substrate, the evolution of the maximal concentration depends on its deformation history (16, 20)

$$c = \frac{c_0}{\sqrt{\tau}} \quad \text{with } \tau = 1 + \frac{4D}{s_0^2}\int_0^t \rho^2 \mathrm{d}t'. \qquad [11]$$

For the two-dimensional stagnation point case considered previously (see Fig. 2) we have

$$\frac{c}{c_0} = \left[1 + \frac{2}{Pe}\left[(\mathrm{e}^{2\gamma t} - 1)\sin^2\theta + (1 - \mathrm{e}^{-2\gamma t})\cos^2\theta\right]\right]^{-1/2}, \qquad [12]$$

with $Pe = \gamma s_0^2/D$ the Péclet number (typically much larger than unity). Note that c is minimal along the stretching direction $\theta = \pi/2$ and maximal along the compression one $\theta = 0$. In the large time limit $\gamma t \gg 1$

$$c \simeq \frac{c_0\sqrt{Pe}\ \mathrm{e}^{-\gamma t}}{\sqrt{2}\sin\theta}. \qquad [13]$$

The stronger a lamella is stretched, the faster its concentration decays, while its width remains blocked at the Batchelor scale (21) $s \sim \sqrt{D/\lambda}$.

In order to compute the concentration distribution $p(c)$ in a mixture, one has to account for the fact that long lamellae weigh the final distribution more than shorter ones. Hence, $p(c)$ must be weighed by the area $\mathcal{A} = s\sqrt{\tau}\ell$ and scalar conservation $\mathcal{A} \times c = \mathcal{A}_0 \times c_0$ leads to

$$p(c)\mathrm{d}c = \frac{\mathcal{A}\ q(\lambda)\mathrm{d}\lambda}{\mathcal{A}_{\mathrm{tot}}} = \frac{\mathcal{A}_0 c_0}{\mathcal{A}_{\mathrm{tot}} c} h(\theta)\mathrm{d}\theta, \qquad [14]$$

in a diluting medium of total area $\mathcal{A}_{\mathrm{tot}}$. Differentiating Eq. (13) with respect to θ and considering the directions close to compression ($\theta \simeq 0$ or $\cos\theta \simeq 1$),

$$p(c) \simeq \frac{c_{\min} c_0}{c^3}, \qquad [15]$$

with $c_{\min} \simeq c_0 \mathrm{e}^{-\gamma t}\sqrt{Pe/2}$ the concentration of a lamella aligned with the fast stretching direction $\theta = \pi/2$. Fig. 5(a) shows that the exact p.d.f. of concentration indeed scale algebraically as c^{-3} above the minimal concentration $c_{\min}$, which decays exponentially in time. The reason is that, when lamellae are oriented at random, some of them fall in a 'reservoir' close to the 'compression' direction for which diffusion is severely slowed. This reservoir shrinks in time, but remains there: Eq. (13) indicates that the fraction of initial angle (for which $c \sim 1$) decays in time as $\theta \sim e^{-\gamma t}\sqrt{Pe} \sim c_{\min}$. Once these lamellae with concentration c close to 1 reach the mixing time, they contaminate the entire $p(c)$ as they catch-up the 'fast' elongational regions by reorienting in the flow. Although they represent a vanishingly small fraction of the lamellae population, they build the algebraic $p(c) \sim c^{-3}$ concentration distribution. This scenario is essentially insensitive to the nature of the stirring protocol since it simply relies on the presence of a compression direction and thus also holds for a 2D sine flow (Fig. 5(b)). It is independent of the dimension of space as well since the same scaling law c^{-3} is found numerically in a 3D stagnation point and for a turbulent flow (Fig. 5(c)).

Similar results are obtained experimentally for mixing of a fluorescent dye by the chaotic flow inside a cylindrical container rotating around its axis and tilted from the vertical –also called Soft-Mixer (18, 22)–. The long-time concentration distribution in this 3D chaotic flow (the experimental apparatus and details of the measurements are described in (23)) present a clear c^{-3} tail, distinct from the low concentration background fluctuations (Fig. 6). The concomitant presence of reorienting, compressing scalar lamellae, carrying a concentration much larger than average, is clearly visible in Fig. 1(c).

Finally, by contrast with the lognormal model for which substrate incompressibility constraints the ratio of the mean to the variance of the elongations to a value which depends on the dimension of space (6, 7), incompressibility is here built into the lamellae dynamics, giving rise to the exponential tails of $q(\lambda)$ and the algebraic tails of the concentration distribution $p(c) \sim c^{-\beta}$. Importantly, this *geometric* constraint imposes $\beta = 3$ exactly, independent of the space dimension, a fact at contrast with the coincidental, and fundamentally different result $\beta \approx 3$ from the lognormal route, or for locally controlled flows (24) in two-dimensions.

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Conclusion

Incompressibility induces spatial correlations in fluid flows (25). We have shown here how rare compression events are crucially important in building the universal long-time, low-stretch statistics of the Lyapunov exponents, hence the ubiquitous c^{-3} concentration distribution in stirred mixtures. The phenomenon is obvious from a simple kinematic construction on a stagnation point flow, and is completely independent of the lognormal cascade of stretchings usually invoked in this context. Although becoming increasingly rare, these events entirely rule the long-time statistical content of stirred mixtures, whether chaotic or turbulent.

This conclusion remains valid until the diffusive overlaps between the lamellae change the paradigm of the mixture route towards uniformity, which then is of a self-convolution nature (26).

Exact formulae

Analytic formulae for the two-dimensional stagnation point. We here derive exact formulae for the distribution of Lyapunov exponents and concentration levels in a two-dimensional stagnation point $\boldsymbol{u} = {}^T[-\gamma x, \gamma y]$. With $\boldsymbol{\ell}_0 = {}^T[\ell_0 \cos(\theta), \ell_0 \sin(\theta)]$, we get

$$\lambda = \frac{1}{2t} \ln \left[\mathrm{e}^{-2\gamma t} \cos^2\theta + \mathrm{e}^{2\gamma t} \sin^2\theta \right] \qquad [16]$$

Sampling initial conditions uniformly in $[0, \pi/2]$, we get $q(\lambda) = 2/(\pi \partial_\theta \lambda)$. Using $\sin^2\theta = (\mathrm{e}^{2\lambda t} - \mathrm{e}^{-2\gamma t})/(\mathrm{e}^{2\gamma t} - \mathrm{e}^{-2\gamma t})$, we get $2\cos(\theta)\sin(\theta)\mathrm{d}\theta = 2t\mathrm{e}^{2\lambda t}\mathrm{d}\lambda/(\mathrm{e}^{2\gamma t} - \mathrm{e}^{-2\gamma t})$. Using $\cos^2(\theta) = (\mathrm{e}^{2\gamma t} - \mathrm{e}^{2\lambda t})/(\mathrm{e}^{2\gamma t} - \mathrm{e}^{-2\gamma t})$, we obtain

$$q(\lambda) = \frac{2t\mathrm{e}^{2\lambda t}}{\pi \sqrt{\mathrm{e}^{2\gamma t} - \mathrm{e}^{2\lambda t}} \sqrt{\mathrm{e}^{2\lambda t} - \mathrm{e}^{-2\gamma t}}}. \qquad [17]$$

This distribution is plotted in the inset in Fig. 2 (thick coloured lines). The corresponding distribution of elongations $Q(\rho) \sim \mathrm{e}^{-\gamma t}$ is uniform, and is consistent with the incompressibility constraint

$$\int_{-\infty}^{+\infty} \frac{Q(\rho)}{\rho^2} \mathrm{d}\rho = \int_{\mathrm{e}^{-\gamma t}}^{\mathrm{e}^{\gamma t}} \frac{Q(\rho)}{\rho^2} \mathrm{d}\rho = 1 \qquad [18]$$

[Note that in d dimensions, incompressibility imposes $\int \rho^{-d} Q(\rho) \mathrm{d}\rho = 1$ or, equivalently, $\int \mathrm{e}^{-\lambda t} q(\lambda) \mathrm{d}\lambda = 1$.] As mentioned in Eq. (13), the maximum concentration levels along lamellae satisfy

$$c = c_0 \left[\eta'(1 - \sin^2\theta) + \eta \sin^2\theta \right]^{-1/2}. \qquad [19]$$

with $\eta' = 1 + 2(1 - \mathrm{e}^{-2\gamma t})/Pe$ and $\eta = 1 + 2(\mathrm{e}^{2\gamma t} - 1)/Pe$. From Eq. (14), the distribution of maximum concentration satisfies

$$p(c) = \frac{2\mathcal{A}_0 c_0}{\pi \mathcal{A}_{\mathrm{tot}} c \left| \frac{\mathrm{d}c}{\mathrm{d}\theta} \right|} = \frac{2\mathcal{A}_0 c_0}{\pi \mathcal{A}_{\mathrm{tot}} c^4 [\eta - \eta'] \sin\theta \cos\theta}. \qquad [20]$$

To this p.d.f. should be added a Dirac delta $\delta(c)$ of initial amplitude $1 - \mathcal{A}_0/\mathcal{A}_{\mathrm{tot}}$ accounting for the diluting background at $c = 0$. This term, which is not written here for simplicity ensures the normalisation of $p(c)$. Then, from $\sin\theta = \sqrt{(c^{-2} - \eta')/(\eta - \eta')}$ and $\cos\theta = \sqrt{1 - \sin^2\theta}$, we get

$$p(c) \sim \frac{2\mathcal{A}_0 c_0}{\pi \mathcal{A}_{\mathrm{tot}} c^4 \sqrt{c^{-2} - \eta'} \sqrt{\eta - c^{-2}}}. \qquad [21]$$

valid for $1/\sqrt{\eta} < c < 1/\sqrt{\eta'}$. Denoting $\mathrm{e}^{-\gamma t}\sqrt{Pe/2} = c_{\mathrm{min}}$ the maximal concentration of the lamellae aligned in the stretching direction $\theta = \pi/2$, we have in the long-time limit $\gamma t \gg 1$,

$$p(c) \underset{\gamma t \gg 1}{\sim} \frac{2\mathcal{A}_0 c_0}{\pi \mathcal{A}_{\mathrm{tot}} c^3 \sqrt{1 - c^2(1 + 2/Pe)}} \times \frac{c_{\mathrm{min}}}{\sqrt{1 + (1 + 2/Pe)c_{\mathrm{min}}^2 - (c_{\mathrm{min}}/c)^2}} \qquad [22]$$

$$\underset{\gamma t \gg 1, Pe \gg 1}{\sim} \frac{2\mathcal{A}_0 c_0 c_{\mathrm{min}}}{\pi \mathcal{A}_{\mathrm{tot}} c^3 \sqrt{1 - c^2} \sqrt{1 + c_{\mathrm{min}}^2 - (c_{\mathrm{min}}/c)^2}}. \qquad [23]$$

These exact distributions are plotted in Fig. 5(a) as colored lines with $\mathcal{A}_0/\mathcal{A}_{\mathrm{tot}} = 10^{-3}$ and $c_0 = 1$. For concentration levels c far from the extremes in the interval $[c_{\mathrm{min}}, 1]$, we have a fair estimate $p(c) \simeq c_0 c_{\mathrm{min}}/c^3$ as in Eq. (15) of the main text. The mean squared concentration $\langle c^2 \rangle = \int c^2 p(c) \mathrm{d}c$ can be calculated by splitting the integral at $c = \sqrt{c_{\mathrm{min}}}$, neglecting the term $\sqrt{1 - c^2}$ for $c < \sqrt{c_{\mathrm{min}}}$ and neglecting the term $\sqrt{1 + c_{\mathrm{min}}^2 - (c_{\mathrm{min}}/c)^2}$ for $c > \sqrt{c_{\mathrm{min}}}$. It leads for small c_{min} to

$$\langle c^2 \rangle = \frac{\mathcal{A}_0 c_0^2 \sqrt{2Pe}}{\pi \mathcal{A}_{\mathrm{tot}}} \mathrm{e}^{-\gamma t} \times \left[\gamma t - \frac{1}{2} \ln \left(\frac{Pe}{32} \right) \right]. \qquad [24]$$

If all lamellae were aligned in the stretching direction, the mean squared concentration would decay as $\langle c^2 \rangle \sim \mathrm{e}^{-\gamma t}$.

We have conducted the discussion with the maximal concentration c carried by each lamella, whose transverse spatial profile is Gaussian, translating into a $\cup$-shaped concentration distribution of C parameterized by c (16). The full concentration distribution $P(C)$ when $p(c)$ is the distribution of maximal concentrations is

$$P(C) = \int_{1/\sqrt{\eta}}^{1/\sqrt{\eta'}} \frac{p(c)}{C \sqrt{\ln(c/C)}} \mathrm{d}c. \qquad [25]$$

This distribution is plotted in Fig. 5(a) and is very close to $p(C)$ for $C > c_{\mathrm{min}}$. The mean squared concentration is simply given by $\langle C^2 \rangle = \langle c^2 \rangle \sqrt{\pi/2}$.

Analytic formulae for the three-dimensional stagnation point. In that case, the eigenvalues of ${}^T\!YY$ are $\mathrm{e}^{-2\gamma t}$, 0 and $\mathrm{e}^{2\gamma t}$. Using formula (2.46) from (17), we have, for $1 < \rho < \mathrm{e}^{\gamma t}$,

$$Q(\rho) = \frac{2\mathrm{e}^{-\gamma t}}{\pi \sqrt{1 - \rho^{-2}} \sqrt{1 - \mathrm{e}^{-4\gamma t}}} \times K \left[-\frac{(1 - \rho^2 \mathrm{e}^{-2\gamma t})(1 - \mathrm{e}^{-2\gamma t})}{(\rho^2 - 1)(1 - \mathrm{e}^{-4\gamma t})} \right]. \qquad [26]$$

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Here $K(\cdot)$ is the complete elliptic integral of the first kind (27). For $\mathrm{e}^{-\gamma t} < \rho < 1$, we have

$$Q(\rho) = \frac{2\rho \mathrm{e}^{-\gamma t}}{\pi\sqrt{1-\rho^2}\sqrt{1-\mathrm{e}^{-4\gamma t}}} \times K\left[-\rho^2 \frac{(1-\rho^{-2}\mathrm{e}^{-2\gamma t})(1-\mathrm{e}^{-2\gamma t})}{(1-\rho^2)(1-\mathrm{e}^{-4\gamma t})}\right]. \quad [27]$$

Hence, in the long time limit $\gamma t \gg 1$, we get

$$Q(\rho) \simeq \frac{2}{\pi} K(-\rho^{-2})\mathrm{e}^{-\gamma t} \simeq \mathrm{e}^{-\gamma t}, \text{ for } 1 \ll \rho \ll \mathrm{e}^{\gamma t},$$
$$\text{and} \quad Q(\rho) \simeq \frac{2}{\pi} K(-\rho^2)\rho \mathrm{e}^{-\gamma t} \simeq \rho \mathrm{e}^{-\gamma t}, \text{ for } \mathrm{e}^{-\gamma t} \ll \rho \ll 1. \quad [28]$$

Incompressibility is ensured by the lowest branch of the elongation distribution since

$$\int_{\mathrm{e}^{-\gamma t}}^{1} \frac{Q(\rho)}{\rho^3}\mathrm{d}\rho \xrightarrow[\gamma t \gg 1]{} 1 \quad \text{and} \quad \int_{1}^{\mathrm{e}^{\gamma t}} \frac{Q(\rho)}{\rho^3}\mathrm{d}\rho \xrightarrow[\gamma t \gg 1]{} 0 \quad [29]$$

Equivalently,

$$q(\lambda) \simeq t\mathrm{e}^{\lambda t - \gamma t}, \text{ for } -\gamma \ll \lambda \ll 0,$$
$$\text{and} \quad q(\lambda) \simeq t\mathrm{e}^{2\lambda t - \gamma t}, \text{ for } 0 \ll \lambda \ll \gamma, \quad [30]$$

as plotted in Fig. 3.

Materials and Methods

GSO method. In order to compute the distribution of elongations and Lyapunov exponents, the eigenvalues $\rho_1, \cdots, \rho_d$ of ${}^TYY$ are needed, where $\dot{Y} = JY$, $Y(t=0) = I_d$ and $J = \nabla \boldsymbol{u}$ for the elongation of lines and $J = -{}^T\nabla \boldsymbol{u}$ for the elongation of surfaces. In this paper, we consider conservative systems and, hence, require that, at all times, $\rho_1 \cdots \rho_d = 1$. Furthermore, we need the small eigenvalues to be not polluted numerically by the large eigenvalues. This constraint was satisfied numerically following a method developed in (3). We summarise the steps here. First, we consider a time dependent orthonormal basis $\boldsymbol{e}_1, \cdots, \boldsymbol{e}_d$ such that

$$\dot{\boldsymbol{e}_1} = J\boldsymbol{e}_1 - J_{11}\boldsymbol{e}_1,$$
$$\dot{\boldsymbol{e}_2} = J\boldsymbol{e}_2 - J_{22}\boldsymbol{e}_2 - (J_{12}+J_{21})\boldsymbol{e}_1, \quad [31]$$
$$\boldsymbol{e}_3 = \boldsymbol{e}_1 \wedge \boldsymbol{e}_2,$$

in three dimensions $d = 3$. Here, $J_{ij} = {}^T\boldsymbol{e}_i J \boldsymbol{e}_j$ are the components of J in the basis $(\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3)$ (but not in the fixed initial basis (x, y, z)). This set of equations is numerically solved using the trapezoidal rule. Second, the differential equation

$$\dot{D} = \begin{bmatrix} J_{11} & J_{12}+J_{21} & J_{31}+J_{13} \\ 0 & J_{22} & J_{32}+J_{23} \\ 0 & 0 & J_{33} \end{bmatrix} D \quad [32]$$

is solved using the modified second-order Runge-Kutta method from (8). Third, the eigenvalues of the symmetric matrix ${}^TDD$ are computed using the appropriate routine from the floating-point arithmetic with arbitrary precision Python library mpmath (28). Noting that ${}^TYY = {}^TDD$, the result of this procedure gives ρ_1, ρ_2 and ρ_3. Following this method, we managed to satisfy $\rho_1\rho_2\rho_3 = 1$ within a 10^{-3} precision at all times.

ACKNOWLEDGMENTS. We are indebted to William R. Young who, pointed out to us, during a 2023 Cargèse summer school on Mixing, the shortcomings of the lognormal model of elongations to interpret concentration distributions.

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